PAIR OF STRAIGHT LINE

- **Second Degree Homogeneous Equation**: A second degree homogeneous equation is $ax^2 + 2hxy + by^2 = 0$ ------→(1)

  (a) If $h^2 > ab$ then (1) will represent a pair of straight lines passing through.

  (b) If $h^2 = ab$ then (1) will represent a pair of identical straight lines passing through.

  (c) If $h^2 < ab$ then (1) will represent the point 'origin'.

  (d) Slopes of the lines given by (1) are given by $m_1 = \dfrac{-h + \sqrt{h^2 - ab}}{b}$, $m_2 = \dfrac{-h - \sqrt{h^2 - ab}}{b}$.

  (e) $m_1 + m_2 = \dfrac{-2h}{b}$.

  (f) $m_1 m_2 = \dfrac{a}{b}$.

  (g) Lines are given by $y = m_1 x$ and $y = m_2 x$.

  (h) Angle between the lines by (1) is $\tan^{-} = \left|\dfrac{2\sqrt{h^2 - ab}}{a + b}\right|$.

  (i) If $a + b = 0$, then lines are perpendicular to each other.

- **Identification of Curves**: General equation of second degree is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ---------------→(1) and

  $$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2.$$

  (a) If $\Delta = 0$ and $h^2 > ab$, then (1) represents intersecting lines.

  (b) If $\Delta = 0$ and $h^2 = ab$, then (1) represents pair of parallel lines or coincident lines and distance between these pair of parallel lines is $= 2 \times \sqrt{\dfrac{g^2 - ac}{a(a+b)}} = 2 \times \sqrt{\dfrac{f^2 - bc}{b(a+b)}}$.

  (c) If $\Delta = 0$ and $h^2 < ab$, then (1) represents a point only given by $\left(\dfrac{hf - bg}{ab - h^2}, \dfrac{hg - af}{ab - h^2}\right)$.

  (d) If $\Delta \neq 0$, $a = b$ and $h = 0$, then (1) represents a circle.

  (e) If $\Delta \neq 0$ and $h^2 > ab$, then (1) represents a hyperbola.

  (f) If $\Delta \neq 0$ and $h^2 = ab$, then (1) represents a parabola.

  (g) If $\Delta \neq 0$ and $h^2 < ab$, then (1) represents an ellipse.

(j) Slopes of the lines given by (1) are given by $m_1 = \frac{-h+\sqrt{h^2-ab}}{b}$, $m_2 = \frac{-h-\sqrt{h^2-ab}}{b}$.

(k) $m_1 + m_2 = \frac{-2h}{b}$.

(l) $m_1 m_2 = \frac{a}{b}$.

(m) Angle between the lines by (1) is $\tan^{-} = \left|\frac{2\sqrt{h^2-ab}}{a+b}\right|$.

(h) If $a+b=0$, then lines are perpendicular to each other.

(i) Point of intersection of lines given by (1) is $x_1 = \frac{hf-bg}{ab-h^2}$, $y_1 = \frac{hg-af}{ab-h^2}$.

(j) Equation of angle bisector of angles formed by (1) are given by $\frac{(x-x_1)^2-(y-y_1)^2}{a-b} = \frac{(x-x_1)(y-y_1)}{h}$.

- Joint equation of pair of straight lines passing through origin and point of intersection of $ax^2+2hxy+by^2+2gx+2fy+c=0$ ------→(1) and $lx+my+n=0$ -------→(2) are given by $ax^2+2hxy+by^2+(2gx+2fy)\left(-\frac{lx+my}{n}\right)+c\left(-\frac{lx+my}{n}\right)^2=0$.

****